Retail Sales Performance Analysis: Understanding the Impact of Economic and Seasonal Factors

PROJECT PROPOSAL FOR TABLEAU PROJECT:

Executive Summary

This project analyzes retail store sales performance using historical sales data combined with economic and environmental factors. The objective is to understand how external factors such as holidays, fuel prices, temperature, and unemployment influence weekly sales across stores and departments. I selected this dataset because retail analytics is a real-world business problem where data visualization helps managers quickly identify sales patterns, seasonal trends, and high-performing stores to support strategic decision-making.

Why (Project Objective / Business Case)

Retail companies must constantly monitor store performance and external economic conditions to optimize sales and marketing strategies.

The goals of this project are:

1. Identify **sales trends across stores and departments over time**.
2. Understand the **impact of holidays and promotions on weekly sales**.
3. Examine whether **economic indicators (CPI, unemployment, fuel prices)** affect retail sales.
4. Identify **top-performing stores and departments**.
5. Provide actionable insights that help management **improve sales planning and promotional strategies**.

Expected outcomes:

- A Tableau dashboard summarizing store performance
- Identification of seasonal patterns and high-impact events
- Data-driven recommendations for sales planning

Who (Audience / Stakeholders)

The visualization is designed primarily for **retail business decision-makers and analysts**.

Primary Audience

- Retail Operations Managers
- Sales Analysts
- Regional Store Managers

These stakeholders need quick insights into sales performance and external factors influencing demand.

Persona Document

Persona 1 — Retail Operations Manager

Name: Sarah Mitchell
Age: 42
Gender: Female

Representative Photo:

Goals

- Monitor overall sales performance across stores
- Identify which stores or departments need attention
- Understand how promotions and holidays affect sales

Challenges / Needs

- Large datasets are difficult to interpret quickly
- Needs a dashboard that highlights trends and anomalies
- Needs quick comparisons between stores

Context of Use

Sarah views the dashboard during **weekly performance review meetings** to monitor store performance and decide on promotional strategies.

Persona 2 — Sales Data Analyst

Name: Daniel Watson
Age: 30
Gender: Male

Representative Photo:

Goals

- Analyze trends and correlations in sales data
- Investigate external factors influencing sales performance
- Identify patterns for future sales forecasting

Challenges / Needs

- Needs interactive visualizations to explore relationships between variables
- Requires the ability to filter by store, department, and time

Context of Use

Daniel uses the dashboard to **perform exploratory analysis and present insights to leadership teams**.

What (Dataset Selection)

Dataset: Retail Data Analytics Dataset

Data source: Kaggle.com

This dataset contains historical sales data from **45 retail stores**, including weekly sales, store information, economic indicators, and promotional markdowns.

Key variables include:

- Store ID
- Department ID
- Date
- Weekly Sales
- Temperature
- Fuel Price
- CPI (Consumer Price Index)
- Unemployment Rate
- Holiday Indicator
- Promotional markdowns

Reasons for selecting this dataset:

- Real-world retail business context
- Includes **date fields suitable for time-series analysis**
- Contains **multiple numerical and categorical variables**
- Enables exploration of **economic factors affecting sales**
- Ideal for creating **interactive dashboards and storytelling visualizations**

How (Proposed Visualization Approach)

The final Tableau dashboard will consist of multiple visualizations designed to tell a clear data story.

Proposed visualizations:

Sales Performance Dashboard

- KPI cards (Total Sales, Avg Weekly Sales)
- Sales trend line over time

Store Comparison Dashboard

• Bar charts comparing **sales by store and department**

External Factors Dashboard

• Scatter plots showing relationships between

- Fuel price vs sales
- Unemployment vs sales
- Temperature vs sales

Seasonal Analysis

• Line charts showing **holiday vs non-holiday sales**
• Heatmaps of **weekly sales by store**

Interactive Features

• Store filter
• Department filter
• Date range filter

• Holiday toggle

These visualizations will allow stakeholders to **quickly explore patterns and identify opportunities for sales improvement**.

Challenges

Expected Challenges

1. **Data Cleaning**
    a. Handling missing values in markdown columns
    b. Ensuring dates are correctly formatted for time-series analysis
2. **Large Dataset Size**
    a. The dataset contains hundreds of thousands of rows, which may need sampling for faster Tableau performance.
3. **Interpreting Economic Indicators**
    a. Understanding how macroeconomic factors influence retail sales.
4. **Designing Effective Dashboards**
    a. Balancing visual clarity with the amount of information displayed.

Areas I Hope to Gain Experience

- Advanced Tableau dashboard design
- Interactive visualization techniques
- Time-series analysis in Tableau
- Storytelling with business data

Questions for Peers

1. What visualizations are most effective for showing the impact of external economic factors on sales?
2. Would a **heatmap or line chart** better highlight seasonal trends?
3. How many dashboards should be included to maintain clarity without overwhelming users?

Data Cleaning and Preparation Steps

Before importing the dataset into Tableau, several data cleaning and preparation steps were performed to ensure the dataset was structured appropriately for analysis and visualization.

First, the dataset was downloaded from the Kaggle datasets repository. The original data consisted of three related CSV files containing sales data, store information, and economic indicators.

Using Python and the pandas library, the three datasets were merged into a single dataframe based on their common fields. This allowed all relevant stores, sales, and economic information to be analyzed together.

The original sales dataset contained more than 420,000 rows. To improve performance and ensure faster visualization in Tableau Public, a random sample of 20,000 rows was selected from the merged dataset while preserving the overall distribution of the data.

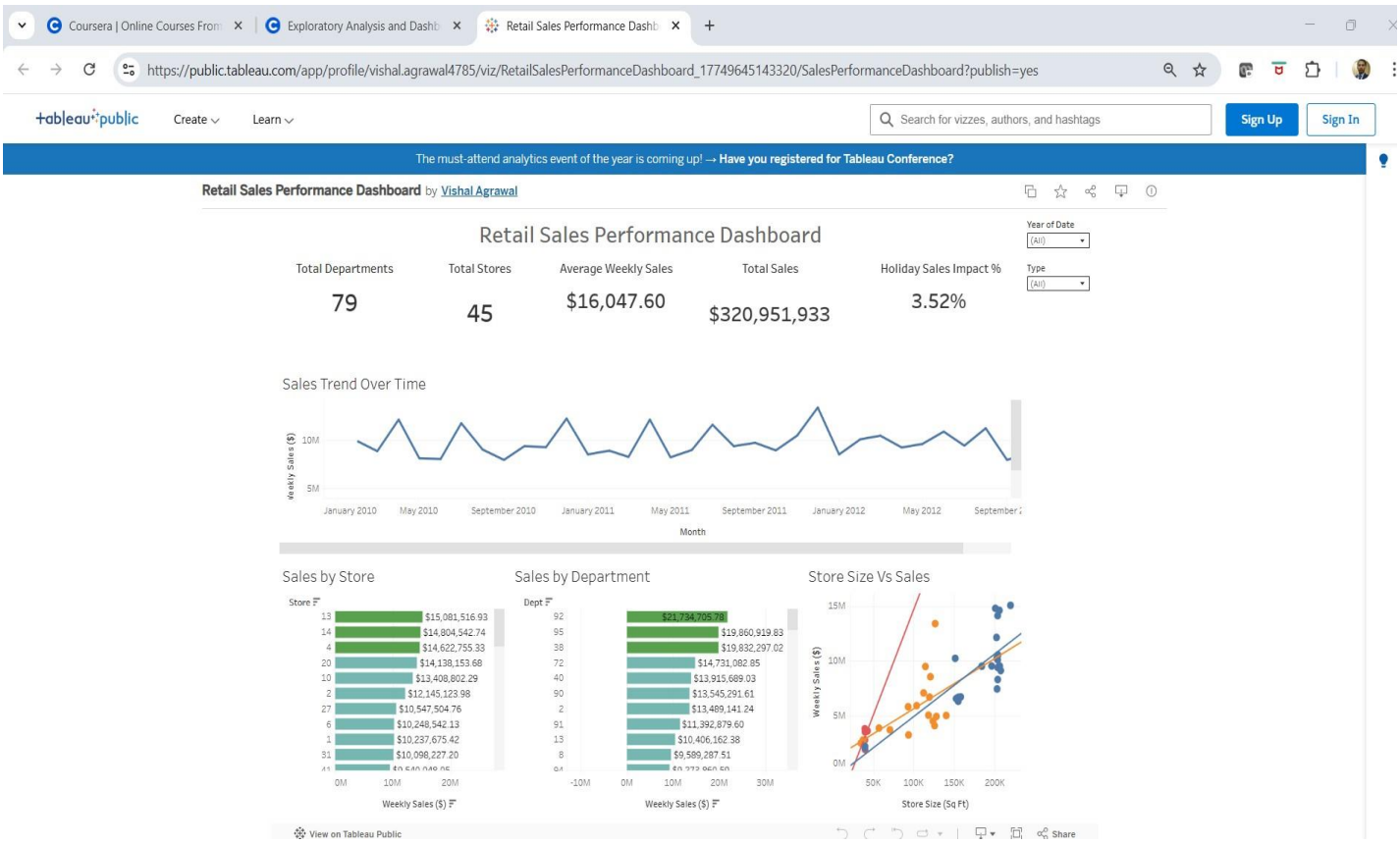
The **Date** column was converted to a datetime format in pandas to ensure proper handling of time-based analysis in Tableau.

Several of the **MarkDown** columns contained missing values. Since these fields represent promotional discounts that were not applied in many cases, the missing values were replaced with 0 to indicate the absence of markdown promotions.

Additionally, two new columns, **Month** and **Year**, were created from the Date field to simplify filtering, grouping, and aggregation when building visualizations in Tableau.

After completing these preparation steps, the cleaned dataset was exported as a CSV file and imported into Tableau Public for further analysis and visualization.

Preliminary Dashboard Image:



Dashboard Preparation questions:

1. Why did you choose these KPIs? What insights do they provide?

I selected **Total Sales, Average Weekly Sales, Total Stores, Total Departments, and Holiday Sales Impact %** as the key performance indicators.

These KPIs provide a quick overview of the overall business performance. **Total Stores and Total Departments** indicate the scale of the dataset being analyzed. **Total Sales** shows the overall revenue generated across all stores, while **Average Weekly Sales** provides an estimate of the typical weekly performance of the retail network.

The **Holiday Sales Impact %** KPI highlights how sales change during holiday periods compared to non-holiday weeks. This helps determine whether holidays significantly influence retail sales performance.

2. How did you apply visual design principles (color, shape, size, layout)?

I applied several visual design principles to improve the clarity and usability of the dashboard.

Color was used as a primary pre-attentive attribute to help users quickly identify patterns. For example, stores and departments were color-coded to highlight performance groups such as **Top 3 performers, above-average performers, and below-average performers**.

Size was used in the KPI cards by increasing the font size of the metric values, allowing users to immediately notice the most important metrics on the dashboard.

The **layout** was structured to guide the user's attention logically. The **top row contains KPIs** for a quick summary, the **middle section contains a sales trend line chart**, and the **bottom row contains comparison charts and scatter plots**. This layout helps users move from high-level performance indicators to more detailed analysis.

3. How does your dashboard use KPI principles (filters, parameters, actions)?

The dashboard includes **interactive filters for Year and Store Type**, allowing users to dynamically analyze sales performance for specific time periods and store categories. When these filters are applied, both the KPI values and the charts update accordingly.

Additionally, a **dashboard highlight action** was implemented. Selecting a store in the **Sales by Store bar chart** highlights the corresponding store in the **Store Size vs Sales scatter plot**, making it easier to compare individual store performance within the overall distribution.

These interactive features enable users to explore the data more effectively and gain deeper insights.

4. What worked—and what didn't—in terms of meeting your audience's needs?

Several parts of the dashboard effectively address the needs of the intended audience.

For example, a **retail store manager** can quickly identify which stores and departments perform best using the ranked **Sales by Store** and **Sales by Department** bar charts. These visualizations make it easy to compare performance across different business units.

A **sales data analyst** can analyze patterns in the data using the **Sales Trend Over Time line chart**, which shows how sales change across months and years. Additionally, the **Store Size vs Sales scatter plot** helps identify the relationship between store size and sales, revealing a positive correlation between these variables.

One limitation of the dashboard is that it does not fully explore the impact of **external factors such as fuel prices, temperature, and economic indicators** due to space constraints. These factors could be further analyzed in an additional dashboard or extended analysis.

5. How long did your exploration take, and what surprised you?

The exploration and dashboard development process took approximately **four hours**.

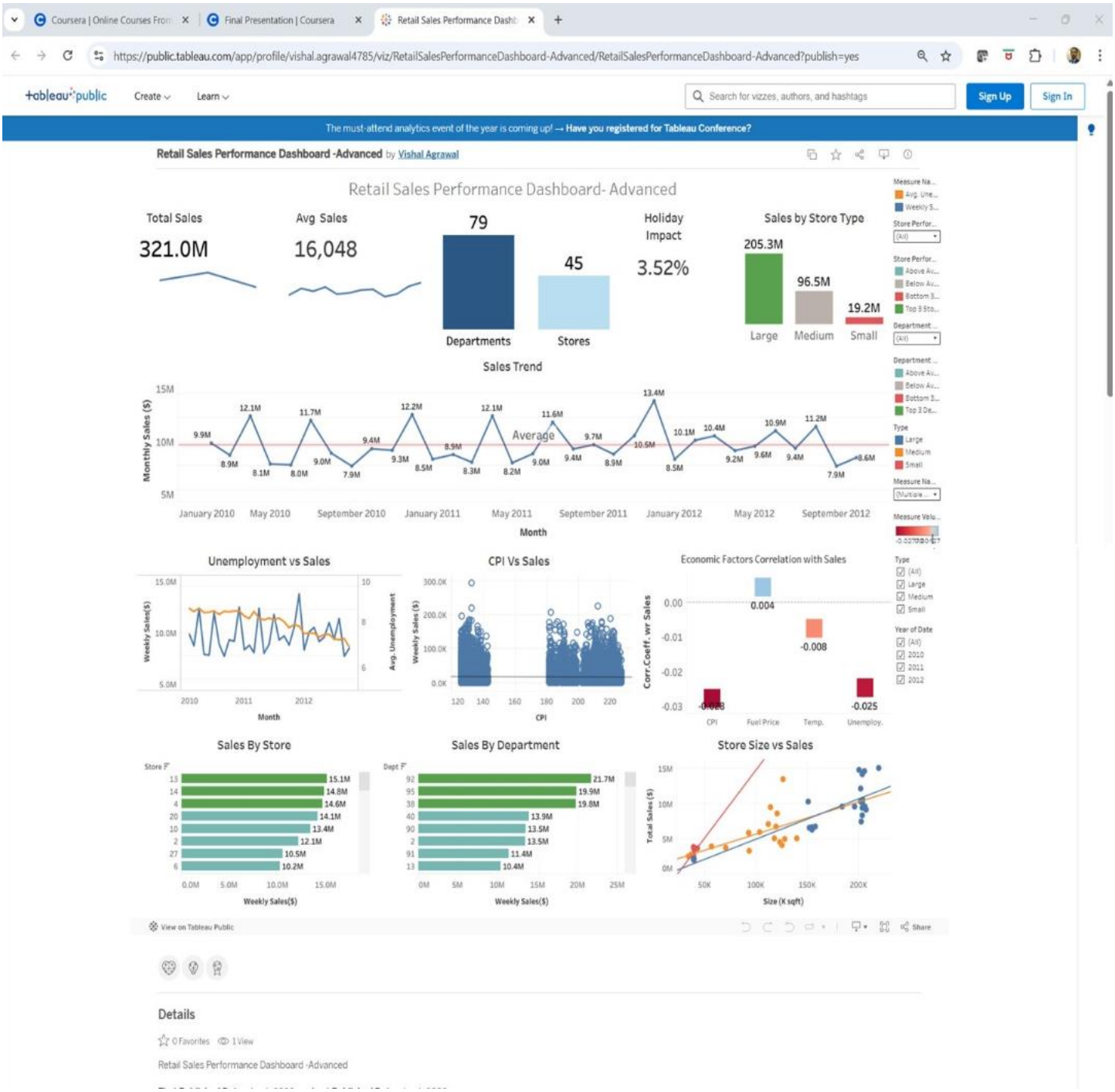
One surprising insight was the **large disparity in sales across stores and departments**, with some stores significantly outperforming others. Additionally, the analysis suggested that **larger stores tend to generate higher sales**, indicating a positive relationship between store size and sales performance.

Update the Dashboard as per below Instructions:

1. **Finalize Your Visualization or Data Story**
    a. Refine your layout, annotations, and design elements to meet the needs of your intended audience.
    b. Apply cognitive and visual design principles (e.g., visual hierarchy, pre-attentive attributes, simplicity, clarity, tone).
2. **Review Your Persona & Proposal**
    a. Revisit the project goals and user needs defined in Milestone 1.
    b. Make sure your final design reflects those needs.

Final Dashboard on Tableau Public:

Link to Tableau Public Dashboard



Question: A short write-up that explains three specific design choices you made, and how each one supports your audience's needs.

Design Choice 1: Interactive Filtering and Use of Color as a Pre-Attentive Attribute

I included two horizontal bar charts displaying total sales across all **stores and departments**. To make performance differences immediately visible, I applied **color as a pre-attentive attribute** to segment stores and departments into four categories:

- **Top 3 performers (Dark Green)**
- **Bottom 3 performers (Red)**
- **Above Average (Light Green)**
- **Below Average (Grey)**

This color scheme allows users to **instantly identify high and low performers without needing to analyze numeric values**.

Additionally, I implemented **interactive filters** that allow users to focus on specific segments such as *Top 3* or *Bottom 3* performers. This interaction enables faster exploration of the dataset and allows managers to quickly isolate areas requiring attention.

This design choice directly supports the needs of the **Retail Operations Manager**, who must:

- Monitor overall sales performance across stores
- Quickly identify stores or departments that require operational attention

By combining color encoding with interactive filtering, the dashboard enables **rapid performance assessment and targeted decision-making**.

Design Choice 2: Trend Analysis of Sales Performance with Reference Line and Labels

To analyze long-term sales performance, I created a **time-series line chart displaying monthly sales over a 36-month period**. This visualization helps users understand how sales evolve over time and detect seasonal patterns.

To provide context for performance evaluation, I added an **average reference line**, allowing users to compare each month's sales value against the overall average. This makes it easier to determine whether a specific month performed above or below typical levels.

To improve readability, the chart includes:

- **Clear numeric labels**
- **Bold formatting for key values**
- **Sales values displayed in millions**

These formatting decisions ensure that important values can be interpreted quickly and accurately.

This design supports the needs of the **Sales Data Analyst**, who aims to:

- Analyze trends and correlations in sales data
- Identify patterns that may support future sales forecasting

The line chart provides a clear overview of sales performance over time, enabling analysts to **identify seasonal effects and potential forecasting signals**.

Design Choice 3: Scatter Plots to explore relationship between variables, Trend Lines

To explore potential drivers of sales performance, I included **scatter plots analyzing the relationship between sales and external variables**, such as **store size and CPI**.

Scatter plots are particularly effective for examining relationships between two continuous variables. I also added **trend lines**, which help users quickly identify the direction and strength of any potential correlation.

In addition, the dashboard includes **interactive filters for store type and year**, allowing users to analyze relationships within specific segments of the data.

This design supports the needs of both stakeholders:

- **Sales Data Analysts**, who require tools to explore relationships between variables
- **Retail Managers**, who need visualizations that quickly highlight patterns or anomalies within large datasets

These scatter plots transform complex datasets into **visual patterns that can be interpreted quickly**, supporting exploratory analysis and data-driven insights.

Question: Please provide a short reflection on how your final project compares to your original project proposal. Did your plan change? What did you notice or learn during the process?

Reflection on the final project compared to my initial project proposal:

My final dashboard closely follows the objectives outlined in my original project proposal. The primary goal of the project was to analyze **sales trends across stores, departments, and time**, which was successfully implemented using a combination of **bar charts, KPI indicators, and a time-series line chart**.

The dashboard effectively highlights **top-performing and underperforming stores and departments**, using color encoding and interactive filtering to allow users to quickly identify areas requiring attention.

Another key objective of the project was to evaluate the impact of **holidays and economic factors such as CPI, fuel prices, and unemployment** on retail sales. This was addressed through several visualizations. A KPI indicator shows the percentage increase in holiday sales compared to non-holiday periods, revealing that sales increased by approximately **3.5% during holidays**.

To analyze macroeconomic influence, I included a **dual-axis line chart comparing unemployment and sales trends**, along with a **correlation heatmap** examining relationships between sales and multiple economic variables. The results indicated that correlations between sales and these economic indicators were relatively weak, with correlation coefficients close to zero.

One limitation that may have affected these results is that the dataset size was reduced from approximately **420,000 rows to 20,000 rows** to improve performance during analysis.

The final objective of the project was to identify **top-performing stores and departments**, which was accomplished using horizontal bar charts combined with segmentation highlighting the **top three and bottom three performers**.

Overall, the structure of the final dashboard remained largely consistent with the original proposal. The main changes involved enhancing the design by:

- Adding **sparklines to KPI cards**
- Including a **correlation heatmap** to better analyze relationships between variables
- Improving the dashboard layout and visual clarity

Through this project, I gained valuable experience in **data visualization design, dashboard development, and data storytelling**. I learned how to design visualizations with the intended audience in mind, apply visual design principles effectively, and transform raw data into insights that support business decision-making.

The project also reinforced the importance of **data cleaning, performance optimization, and clear communication of analytical insights**, all of which are essential skills for effective data analysis.